Total No. of Questions : 8]

SEAT No. :

PE-2520

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T.E. (Computer Engg./ Computer Science)

Internet of Things & Embedded Systems

(2019 Pattern) (Semester - V) (310245 A)(Elective - I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Attempt question Q.1 or Q.2, Q.3 or Q.4, Q. 5 or Q. 6, Q. 7 or Q. 8.***
2) ***Draw neat & labelled diagrams if necessary.***
3) ***Assume suitable data if necessary.***
4) ***Figures to the right side indicate full marks.***

Q1) a) Compare REST and WebSocket communication methods used in IoT **[6]**

b) Explain 4 pillars of IoT system with diagram **[5]**

c) Design the purpose and requirement specification for smart IoT-based Home monitoring system. **[6]**

OR

Q2) a) Discuss exclusive pair communication model in IoT communication. **[6]**

b) Explain SCADA system & it's applications. **[5]**

c) Explain the domain model specification step with diagram. **[6]**

Q3) a) Discuss the standardization efforts taken by different systems for IoT Protocols. **[6]**

b) Outline IP-based protocols. Explain the working of 6LoWPAN **[6]**

c) Discuss the ZigBee architecture with diagram. **[6]**

OR

P.T.O.

Q4) a) Explain the working of RFID with any application **[6]**

b) Discuss the working of Modbus? Compare the efficiency of Modbus Vs KNX protocol. **[6]**

c) Write short note on M2M protocol and its applications. **[6]**

Q5) a) Explain with diagram WAMP (AutoBahn for IoT) **[6]**

b) Discuss the services provided by Cloud. **[6]**

c) Why Is MQTT the is the preferred protocol for IoT? **[6]**

OR

Q6) a) Determine the role of cloud storage models in IoT. **[6]**

b) Show the use of Cloud Computing while implementing the parcel tracking application. **[6]**

c) Explain in short Amazon S3, Amazon EC 2 **[6]**

Q7) a) Illustrate the key elements of the IoT security with suitable examples of each. **[6]**

b) What is Skynet? How it is useful to user and developers **[6]**

c) Discuss vulnerabilities in IoT. **[5]**

OR

Q8) a) What is threat modelling? What are the benefits of threat modeling? **[6]**

b) Infer security challenges for intrusion detection system for smart home. **[6]**

c) Discus the Light weight cryptography **[5]**